# Understanding Probabilistic Causality

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*User prompt: probabilistic causality*

Response: **Probabilistic causality** is a framework in the philosophy of science and statistics that redefines causal relationships using the language of probability rather than strict determinism.

Instead of asserting that cause $C$ *always* produces effect $E$ (deterministic causality: $C \implies E$), probabilistic causality posits that **a cause alters the probability of its effect occurring**:

$$P(E \mid C) > P(E \mid \neg C)$$

*(assuming all relevant background conditions are held constant or accounted for).*

## Core Premises

1. **Abandonment of Determinism:** Classic Newtonian determinism assumes complete knowledge of initial conditions allows exact prediction of outcomes. Probabilistic causality accommodates intrinsic randomness (e.g., quantum mechanics), incomplete knowledge, or complex systems (e.g., epidemiology, economics).
2. **Probability Raising:** The core intuition is that causes raise the likelihood of their effects. For example, smoking ($C$) increases the probability of developing lung cancer ($E$), even though not every smoker gets lung cancer and some non-smokers do.

## Fundamental Challenges & Refinements

Simple probability raising is insufficient to define causality due to three major theoretical hurdles:

### 1. Spurious Correlations (Common Causes)

Two variables may be highly correlated without any direct causal connection if both are driven by a third variable ($A$).

- **Example:** Ice cream sales ($C$) and drowning rates ($E$) are positively correlated. However, buying ice cream does not cause drowning. Summer heat ($A$) causes both.
- **Solution (Screening Off):** $A$ "screens off" $C$ from $E$ if $C$ provides no additional information about $E$ once $A$ is known:

$$P(E \mid C \wedge A) = P(E \mid A)$$

### 2. Causal Asymmetry

Probability correlation is symmetric: if $P(E \mid C) > P(E)$, then $P(C \mid E) > P(C)$. However, causation is directional ($C \to E$, not $E \to C$). Probabilistic causality relies on temporal precedence ($C$ must precede $E$ in time) or structural interventions to establish directionality.

### 3. Preemption and Interference

A cause might actually *decrease* the probability of an outcome in a specific context due to alternative pathways (Simpson's Paradox).

- **Example:** A birth control pill ($C$) might slightly increase thrombosis risk directly, but by preventing pregnancy (which carries a higher risk of thrombosis), it decreases the overall probability of thrombosis in a population sub-group.

## Modern Frameworks

To resolve these challenges, modern probabilistic causality relies on two dominant paradigms:

### A. Pearl's Structural Causal Models (Do-Calculus)

Judea Pearl shifted the focus from passive observation $P(E \mid C)$ to active intervention $P(E \mid \mathrm{do}(C))$.

- **Observational:** $P(E \mid C)$ asks, *"What is the probability of $E$ given that I observe $C$?"*
- **Interventional:** $P(E \mid \mathrm{do}(C))$ asks, *"What is the probability of $E$ if I actively force $C$ to occur?"*

By breaking external dependencies in Directed Acyclic Graphs (DAGs), Pearl's $\mathrm{do}$-operator isolates genuine causal effects from confounding associations.

### B. Suppes & Reichenbach's Common Cause Principle

Hans Reichenbach proposed that if two events $A$ and $B$ are correlated ($P(A \wedge B) > P(A)P(B)$), there is either a direct causal link ($A \to B$ or $B \to A$) or a common cause $C$ that screens them off. Patrick Suppes later formalized this into explicit temporal and probabilistic axioms.

## Key Applications

- **Epidemiology & Medicine:** Evaluating drug efficacy and risk factors where outcome variations occur across diverse populations.
- **Neuroscience & Computational Brain Modeling:** Analyzing functional and effective connectivity (e.g., Granger Causality, Transfer Entropy) in time-series data like fMRI or EEG.
- **Machine Learning & Counterfactual AI:** Moving beyond associative deep learning models toward causal inference, fairness algorithms, and explainable AI (XAI).